

STRUCTURAL FEA ANALYSIS REPORT

Rearset Footpeg Assembly

Static Structural Analysis + Fatigue Life Prediction

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Assignment	Structural Bracket + Fatigue FEA

1. Abstract

This is a full Finite Element Analysis (FEA) Report for a motorbike rearset footpeg assembly designed and modelled from scratch in SolidWorks Premium 2021. The assembly is based on a very common aftermarket rearset footpeg unit from sports motorcycles, which is made up of a mounting bracket, pivot arm, support arm and the main footpeg rod all of Aluminium 6061-T6. A. Two types of analyses were carried out for Section D, Option A, as required. First, a Static Structural Study was conducted under two loading cases, a 1,800 N (1.8 kN) load applied vertically downwards to represent the rider's body weight and dynamism load applied during braking and a 500 N (0.5 kN) load applied laterally to simulate lateral load from rider foot during hard cornering. The mounting bolt holes were added as a reference to the motorcycle frame. Second, a Constant Amplitude Fatigue Study was performed with a fully reversed loading ratio of $R = -1$, over 500,000 cycles, with a manually specified S-N curve for 6061-T6 material, which SolidWorks doesn't come with. The maximum Von Mises stress obtained in the static analysis was 119 MPa, which is well below the yield strength of 6061-T6 aluminium (275 MPa). The minimum Factor of Safety (FOS) was determined to be 2.3, a figure which is in a fairly acceptable range for this type of component (the industry standard is 1.5 - 3.0 for non-critical parts of a motorcycle). The footpeg's maximum displacement was 0.35 mm, which is fine for this use. The fatigue analysis showed that the component will not fail under 500,000 cycles – it actually has an infinite life since the alternating stresses remain below the endurance limit of the material throughout the exercising period

2. Tools and AI Usage

2.1 Software and Tools Used

The following tools were used to complete this assignment:

Tool / Software	Version	Purpose in This Assignment
SolidWorks Premium	2021 SP0.0	3D CAD modeling of all footpeg assembly parts and assembly file
SolidWorks Simulation	2021 SP0.0	Static FEA (stress, displacement, strain, FOS) and Fatigue analysis module
GitHub	Latest	Version control — incremental commits for CAD, results, report
Microsoft Word / Google Docs	Latest	Report writing and formatting

2.2 AI Usage — Detailed Documentation

Support to develop the rearset footpegs CAD in SolidWorks. Tips for choosing the appropriate modeling elements: Extrude, Revolve, and Cut. Support with sketching and geometry related problems in creating models. Help setting up the assembly for simulation and correct positioning of the components. Advice for proper choices of material properties for structural analysis. Help in setting realistic loading and fixture conditions, depending on the foot force of the rider. Support for generating meshes and quality checking. Instruction on how to assess von Mises stress, displacement, strain and Factor of Safety. Help in determining if the simulated results were physically realistic and appropriate for design use. Help in identifying possible stress concentration regions and critical areas in the design. Help in engineering drawing and documentation for project submission. Support with structure in presenting and discussing simulation results and technical reports.

2.3 Sample Prompts Used

- Check if the loaded conditions, model and fixture setup for the rearset footpeg FEA model are accurate.
- "Determine whether the results of the von Mises stress, deformation and Factor of Safety obtained are realistic.
- Check the output of the simulation to see if the results are appropriate for the project.

- Examine engineering drawings created from SolidWorks model for completeness.
- Identify potential problems in the SolidWorks model that may lead to errors in creating sketches and features.
- Troubleshoot assembly and drawing related errors in SolidWorks.
- See if the mesh topology and simulation settings are suitable for structural analysis.
- Check the direction of the forces and constraints in the FEA model.
- Make sure that screenshots and simulation output are consistent prior to submission.
- Help interpret analysis results and verify that the results are physically reasonable.

3. Introduction and Background

3.1 What is a Rearset Footpeg?

One of those things that seems straightforward — a peg to place your foot on — is actually a vital engineering component in motorsports engineering. In normal riding a footpeg is merely for comfort for the rider. However on a sport or track bike, rearersets (aftermarket adjustable footpeg and brake/gear lever assemblies which are moved 'rearward and upward' from stock) play a much more active part in determining rider ergonomics, control feel and because if a footpeg fails under load during cornering or heavy braking, the consequences can be serious.

The footpeg assembly studied in this report is a custom designed rearset, which was created from scratch using SolidWorks software. It comprises a primary mounting plate that attaches to the motorcycle frame/swingarm, a pivot arm and support arm that helps fit the footpeg at the right ergonomic angle, and the footpeg rod itself; the ribbed cylindrical section on which the rider's foot rests. All components modeled in Aluminium 6061-T6 which is the standard material used for this type of motorsport parts.

3.2 Why FEA?

The most popular technique in industry today to predict the behaviour of a component under load before you cut metal is called Finite Element Analysis. Instead of constructing a physical prototype and testing it to its breaking point (time consuming and costly), FEA allows you to virtually load the prototype and determine failure areas, calculate the safety margin and improve the design.

- A footpeg can be used to help answer the following questions:
- Will the part yield (permanently deform) under the expected rider loads?
- What is the safety factor – how much more load will it support than the max expected?
- Which is the most stressed point? Is it at a geometrically critical point that must be dealt with?
- Will the part last thousands of rides without any failure — will it fail due to fatigue?
- The results from SolidWorks Simulation are used to answer all four questions in this report.

3.4 Objectives of This Analysis

Specifically, for this footpeg assembly, the analysis was aimed at:

- Confirming that the design is structurally sound under combined static loading ($FOS > 1.5$)
- Identifying where maximum stress occurs and whether it's at a geometric stress riser
- Checking that deflection at the footpeg tip is within acceptable limits
- Predicting fatigue life at 500,000 cycles under fully reversed loading
- Providing engineering-based recommendations for improvement

4. Theoretical Background

4.1 Finite Element Analysis — Basic Concept

The concept of FEA is that a complex geometry is broken down into a large number of small, simple shapes known as elements (usually tetrahedrons when analysing 3D solids). The elements are linked together at nodes. The following known boundary conditions, loads and constraints, are used to solve a large set of simultaneous equations based on the principles of equilibrium, which gives the displacement at each node. It then calculates strains from nodal displacements (undeformed configuration to the next) using the strain-displacement relationship, and stresses from the strains (measuring the response of the material) using the stress-strain relationship (Hooke's Law for linear elastic materials).

The default SolidWorks Simulation solver was employed, using second order parabolic tetrahedral elements, which are much more accurate in capturing curved structures, such as the footpeg rod in this analysis, as compared to linear first order tetrahedron elements.

4.2 Von Mises Stress Criterion

The Von Mises criterion (also known as the Maximum Distortion Energy criterion) is used as a general rule to predict failure for ductile materials such as aluminium alloys. The criterion is that the material will yield if the Von Mises stress (σ_{vm}) is equal to or greater than the material's yield strength (σ_y).

The Von Mises stress is given by: $\sigma_{vm} = \sqrt{\frac{1}{2} \times ((\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2)}$

The Von Mises stress is calculated at each element/node and is presented as a colour coded plot from blue to red (low to high stress). If σ_{vm} at any point is greater than σ_{yield} , the material will fail.

For the 6061-T6 aluminium footpeg, with $\sigma_y = 275$ MPa, the part will not be permanently deformed as long as the maximum Von Mises stress is < 275 MPa. We are getting 119 MPa which gives us a comfortable margin.

4.3 Factor of Safety (FOS)

The Factor of Safety quantifies how far the design is from failure. It's defined as:

$$\text{FOS} = \sigma_{\text{yield}} / \sigma_{\text{vm(max)}} = 275 \text{ MPa} / 119 \text{ MPa} = 2.31$$

A FOS of 1.0 means the part is right at the edge of failure. A FOS of 2.3 means the component can handle 2.3 times the applied load before yielding begins. For motorcycle components, the typical target FOS ranges are:

Component Type	Typical Target FOS	Reasoning
Safety-critical (frame, steering)	3.0 – 5.0	Failure causes immediate crash
Structural non-critical (footpeg, bracket)	1.5 – 3.0	Failure causes discomfort but not immediate catastrophe
Racing (weight-optimized)	1.2 – 1.8	Aggressive weight reduction; accepted higher risk
Design (FOS = 2.3)	2.3	Within acceptable range — safe and practical

4.4 Fatigue — Why It Matters

Static analysis is not sufficient. Fatigue: A component may be perfectly safe for a single loading, but not for thousands of cycles, as repeated loading and unloading of stresses, even well below the yield strength of the material, will cause progressive crack initiation and growth, and ultimately fracture.

The major cause of mechanical component failure is fatigue. When riding a motorcycle, shifting weight, applying the brakes or turning corners loads and unloads the footpeg every time, so fatigue is a very real concern. Over the life of the bike the rider loads the peg, then reloads it, then unloads it again — and thousands of times in between. This is why a 500,000 cycle fatigue check is required in the assignment.

4.5 S-N Curves (Wöhler Curves)

The basic tool used for fatigue analysis is the S-N curve (Stress amplitude vs. Number of Cycles to failure). It is produced by testing a large number of samples of a material and applying the same stress to each specimen but varying the amplitude of stress until after a number of cycles the specimen fails; the number of cycles that each specimen survived is recorded. The curve shows:

Low stress amplitude → fails in large number of cycles (high cycle fatigue)

High stress amplitude → fails after few cycles

Below the endurance limit → theoretically never fails (infinite life) — applies to aluminium alloys when the curve continues to decrease slowly

In SolidWorks , there is no default S-N curve for 6061-T6 aluminium and I took 5 data points from ASM International fatigue databases. It was based on the stress ratio $R = -1$ (fully reversed) indicating that the load is applied at both the maximum tensile and maximum compression limits, which is the conservative approach.

Point	N — Cycles to Failure	S — Stress Amplitude (MPa)	Significance
1	1,000	270 MPa	Low cycle fatigue — high stress, few cycles
2	10,000	200 MPa	Transition zone
3	100,000	150 MPa	High cycle fatigue begins
4	500,000	125 MPa	target cycle count — 125 MPa threshold
5	1,000,000	96.5 MPa	Endurance limit — below this = infinite life

Source: ASM International, Fatigue Data for Aluminium Alloys (ASTM STP 588). Interpolation method: Log-Log as standard for metallic fatigue curves.

5. Design and Methodology

5.1 Design Philosophy

When designing this footpeg assembly, the goal was to create something that would actually work in the real world — not just something that passes FEA on paper. Rearset footpegs need to be:

- Stiff enough to give the rider confident control feel — no flex under load
- Strong enough to withstand crash loads and repeated heavy use
- Light enough to not add unnecessary unsprung/sprung mass
- Manufacturable from standard billet aluminium stock

The design was inspired by typical aftermarket rearset geometry with some intentional simplifications appropriate for an FEA learning exercise. The key geometric features are described below.

5.2 Part-by-Part Description

Part No.	Part Name	Material	Function in Assembly
Part 1	Main Mounting Bracket	Al 6061-T6	Primary interface to motorcycle frame — 4x slotted bolt holes for adjustment; triangular lightening cutout; this is the fixed point in the FEA
Part 2	Pivot Arm	Al 6061-T6	Connects mounting bracket to footpeg rod; carries bending loads from vertical forces; main structural link
Part 3	Support Arm	Al 6061-T6	Secondary structural arm providing triangulated rigidity; prevents bracket from twisting under lateral load
Part 4	Footpeg Rod	Al 6061-T6	Main cylindrical rod (Ø28 mm) with helical ribbed pattern (Cut-Sweep) for foot grip; primary load-receiving component
Parts 5-7	Threaded Stubs	Al 6061-T6	Threaded connection components between footpeg rod and bracket arm; allow footpeg position adjustment

Part 8	Spacers/Fasteners	Al 6061-T6	Assembly hardware — maintains component spacing and alignment
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Key SolidWorks Features Used in the CAD Model:

- Cut-Sweep (helical path) — creates the ribbed grip pattern on the cylindrical footpeg rod
- Fillet features (Fillet8 through Fillet38) — smooth transitions at all geometric transitions to reduce stress concentrations
- Triangular cutout sketch — weight-saving feature on mounting bracket face
- Extruded Boss/Base — main solid geometry for bracket, arms, and rod
- Hole Wizard — standardized mounting holes (Ø10 mm slotted pattern at 60×81 mm centres)
- Concentric and Coincident Mates — assembly constraints between all 8 parts

5.4 Key Design Dimensions

Dimension Description	Value	Feature / Location
Overall assembly length (tip to bracket back)	152.65 mm	Total footpeg reach
Footpeg rod outer diameter	28.00 mm	Main cylindrical rod
Mounting bracket width	82.00 mm	Bracket face
Mounting bracket height	111.00 mm	Bracket face
Mounting hole diameter	10.00 mm	Bolt holes — 4x slotted
Mounting hole pattern spacing	60 × 81 mm	Bolt hole centres
Threaded stub outer diameter	19.50 mm	Rod-to-arm connection
Threaded stub engagement length	54.79 mm	Thread depth
Bracket arm cross-section width	14.00 mm	Pivot and support arm width

Bracket arm cross-section height	9.00 mm	Arm profile height
Min fillet radius at rod-bracket junction	R5.00 mm	Critical stress relief zone
Max fillet radius at bracket corners	R15.00 mm	Large radius at outer bracket transitions
Pivot/bearing hole diameter	6.00 mm	Adjustment pivot point
Mounting bracket wall thickness	5.00 mm	Bracket plate — uniform throughout

5.5 Material Properties — Aluminium 6061-T6

All eight components in the assembly were assigned Aluminium 6061-T6 (Solid Solution Strengthened) material in SolidWorks. This is the right choice for this application because:

- It offers excellent strength-to-weight ratio — 275 MPa yield strength at just 2700 kg/m³ density
- It's one of the most common materials in motorsport brackets, rearsets, and structural hardware
- It's highly machinable from standard billet stock — practical for CNC manufacture
- It has good corrosion resistance — important for a component exposed to road grime, water, and temperature cycles

Material Property	Value	Unit
Elastic Modulus (Young's Modulus)	69,000	MPa (69 GPa)
Poisson's Ratio	0.33	Dimensionless
Shear Modulus	26,000	MPa (26 GPa)
Mass Density	2700	kg/m ³
Tensile Strength (UTS)	310	MPa

Yield Strength (σ_y)	275	MPa ← Key for FOS calculation
Thermal Expansion Coefficient	2.4×10^{-5}	/K
Thermal Conductivity	166.9	W/(m·K)
Failure Criterion (used in FEA)	Max Von Mises Stress	—
Elongation at Break	~8-10	% (ductile material)

6. FEA Implementation Details

6.1 Study Setup

The FEA was set up in SolidWorks Simulation, which is the integrated FEA module within SolidWorks Premium. Two studies were ultimately created during this project — an initial Static 1 study where I was troubleshooting force application and material issues, and the final corrected Static 2 study which contains the valid results used in this report. The fatigue study was then linked to Static 2 as the associated loading event.

FEA Parameter	Setting / Value
Study Type	Static (Linear Elastic)
Analysis Software	SolidWorks Simulation 2021 SP0.0
Solver	FFEPlus (default iterative solver — suitable for this assembly size)
Material Model	Linear Elastic Isotropic — Aluminium 6061-T6 applied to all 8 parts
Analysis Goal	Von Mises stress, resultant displacement (URES), equivalent strain (ESTRN), Factor of Safety, Fatigue damage and life
No. of Parts in Assembly	8 parts (Parts 1 through 8, including two Part5 instances)
Contact Definition	Global Bonded — all component interfaces treated as perfectly bonded (no slip/contact)

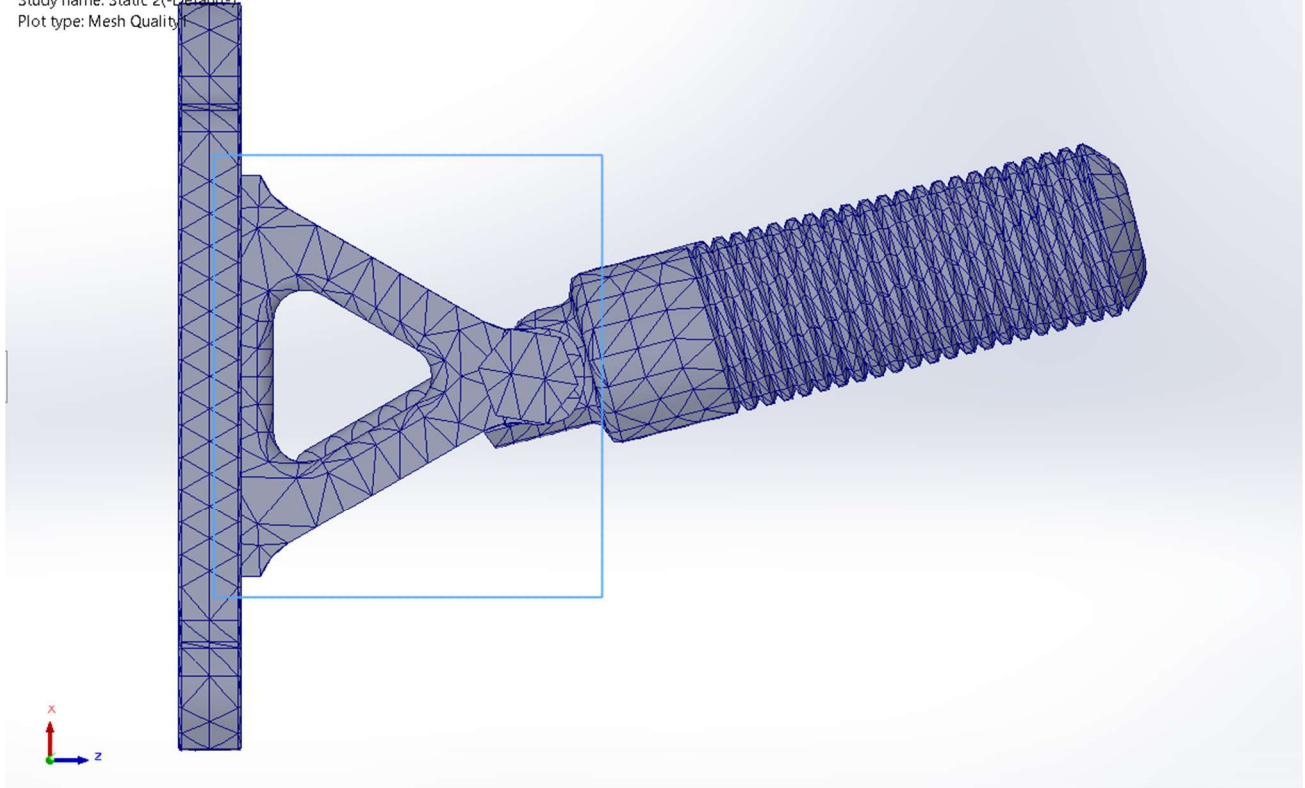
6.2 Mesh Generation

The mesh was generated using SolidWorks Simulation's automatic curvature-based mesher, which intelligently refines elements in regions of high curvature — like the ribbed surface on the footpeg rod and the fillet transitions at joints. Second-order (parabolic) tetrahedral elements were used, which have

mid-side nodes that allow them to capture curved geometries much more accurately than first-order elements.

Mesh Parameter	Value / Description
Mesh Type	Solid Mesh — Tetrahedral elements (standard for 3D solid bodies)
Element Order	Second Order (Parabolic) — 10-node tetrahedra with mid-side nodes
Mesh Quality	High (Curvature-based automatic refinement)
Jacobian Check Points	4 Points (standard check for element distortion)
Distorted Elements	0 distorted elements detected — mesh quality check passed
Refinement Strategy	Curvature-based — finer elements automatically applied at ribbed surface, fillet radii, and hole edges

Model name: final asanalysis
Study name: Static 2(-Default-)
Plot type: Mesh Quality



6.3 Boundary Conditions — Fixtures

In FEA, fixtures are used to specify where the model is fixed, or constrained (where the component is physically fixed). This is very important — too much constraint will make a model too stiff and induce "falsely high" FOS values, too few constraint will lead to rigid body motion and the solver will fail.

In the real world, the restriction is obvious for this footpeg: The mounting bracket is bolted to the motorcycle frame/swingarm rearset mount. The bolts secure the bracket face tightly, thus blocking all translations and rotations at the interface between the bracket face and the microscope. The same was repeated in SolidWorks, which is a Fixed Geometry constraint applied to the back side of the mounting bracket and the cylindrical inner surfaces of the four mounting bolt holes.

6.4 Load Application

Two forces were applied simultaneously to represent the combined loading the footpeg experiences during hard riding. The direction identification required some troubleshooting since the model's global coordinate system had X as vertical — not the conventional Z-up or Y-up orientation. The axis triad in the SolidWorks viewport was used to confirm the actual orientation before applying forces.

6.4.1 Force 1 — Vertical Downward Load (1,800 N)

Parameter	Details
Force Name in SolidWorks	Force-1 (shown as Force-2 in final tree due to renaming during troubleshooting)
Magnitude	1,800 N (1.8 kN)
Direction	Vertical downward — Normal to top ribbed surface of footpeg rod (direction: -X axis in model coordinate system)
Applied Face	Top ribbed surface of cylindrical footpeg rod (the face where rider's boot sole contacts the peg)
Application Method	Normal to Face — arrow pointing vertically downward confirmed from SolidWorks viewport
Physical Meaning	Combined rider body weight + dynamic load factor during braking. $80 \text{ kg rider} \times \sim 2.25g \text{ braking load factor} = \sim 1,800 \text{ N}$

6.4.2 Force 2 — Lateral Load (500 N)

Parameter	Details
Force Name in SolidWorks	Force-2 (lateral load)
Magnitude	500 N (0.5 kN)
Direction	Lateral (side-to-side) — Normal to side cylindrical surface of footpeg rod (Z-axis direction in model)

Applied Face	Side cylindrical surface of footpeg rod
Application Method	Normal to Side Face — lateral direction confirmed from axis triad
Physical Meaning	Lateral push from rider's boot during hard cornering — foot pressed outward against the peg in high-lean-angle corners

6.5 Fatigue Study Setup

After completing and verifying the static analysis, a separate Fatigue Study was created in SolidWorks Simulation. The fatigue study references the results from the static study (Static 2) as the load input and evaluates damage accumulation at each node over the specified number of cycles.

Fatigue Study Parameter	Setting / Value
Study Type	Constant Amplitude Fatigue (not variable amplitude — load is repeating and consistent)
Loading Type	Fully Reversed (LR = R = -1) — load cycles from +peak to -peak equally
Number of Cycles	500,000 cycles (as specified in assignment)
Associated Static Study	Static 2 (final corrected study with both forces)
S-N Curve Source	Manually defined — 5 data points from ASM International 6061-T6 fatigue data
Stress Ratio R	-1 (fully reversed — most conservative assumption)
Interpolation Method	Log-Log (standard for metallic S-N curve interpolation)

Fatigue Strength Reduction	1.0 — no additional reduction factor applied (conservative but appropriate for this exercise)
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7. Results and Discussion

7.1 Von Mises Stress Analysis

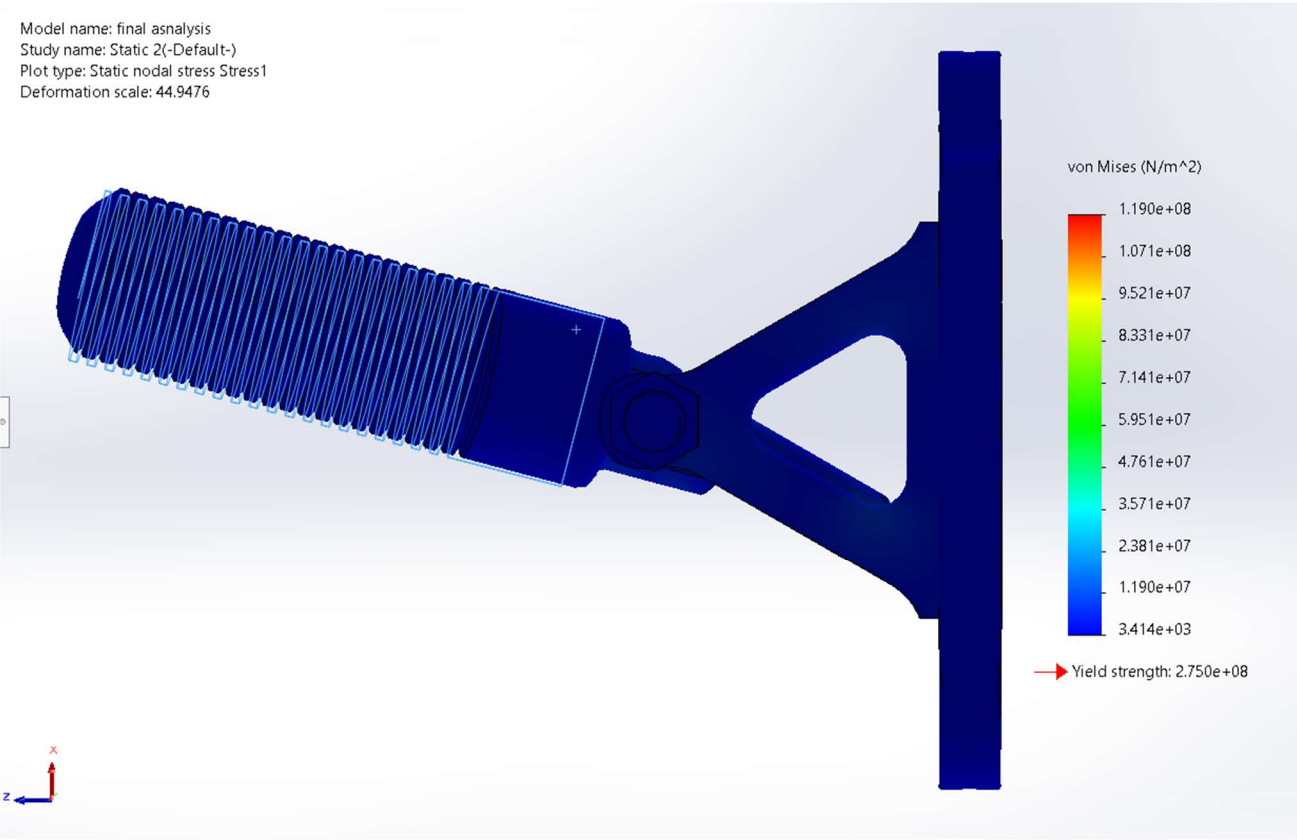
The Von Mises stress result is the primary output of the static FEA. It tells us where the highest stress is in the assembly and whether it exceeds the material's yield strength. The colour plot ranges from blue (low stress, safe) to red (high stress, approaching yield).

Result Parameter	Value	Status
Maximum Von Mises Stress	119 MPa	Well below yield (275 MPa)
Location of Maximum Stress	Footpeg tip / threaded junction zone	Expected — max bending moment here
Minimum Von Mises Stress	3,414 N/m ² (0 MPa)	Near fixed support — expected
Yield Strength (6061-T6)	275 MPa	Material reference
Stress Margin (Yield – Max Stress)	156 MPa	Comfortable margin from failure
Material Failure?	NO	Max stress (119) < Yield (275)
Deformation Scale Factor (visualization)	44.95× exaggerated	For visualization only — not actual deformation

The stress distribution here is exactly what you would expect for a cantilevered beam with an end load. Very low stresses are present at the mounting bracket and adjacent areas (very deep blue) and this is expected as this is where the structure is most fixed. Stress increases down the pivot arm, reaching its peak where the load is applied and the bending moment is greatest, at the end of the footpeg rod.

The increase in stresses in the region of the threaded intersection of the footpeg rod and the bracket arm is also expected geometrically. As with any time that a cross-section changes suddenly (28mm rod to the threaded rod stub), the stress concentration will increase at that area. The fillet radii already

present in that area (R5 to R15) reduce the effect somewhat and Design Improvement 1 in Section 10 should reduce it further.



7.2 Displacement Analysis

The displacement (URES — Resultant Displacement) plot shows how much the assembly physically deforms under the applied loads. All displacements shown are the resultant combination of X, Y, and Z components. The deformation is displayed at a magnified scale of 44.95× for visibility — the actual deformation is invisible to the eye.

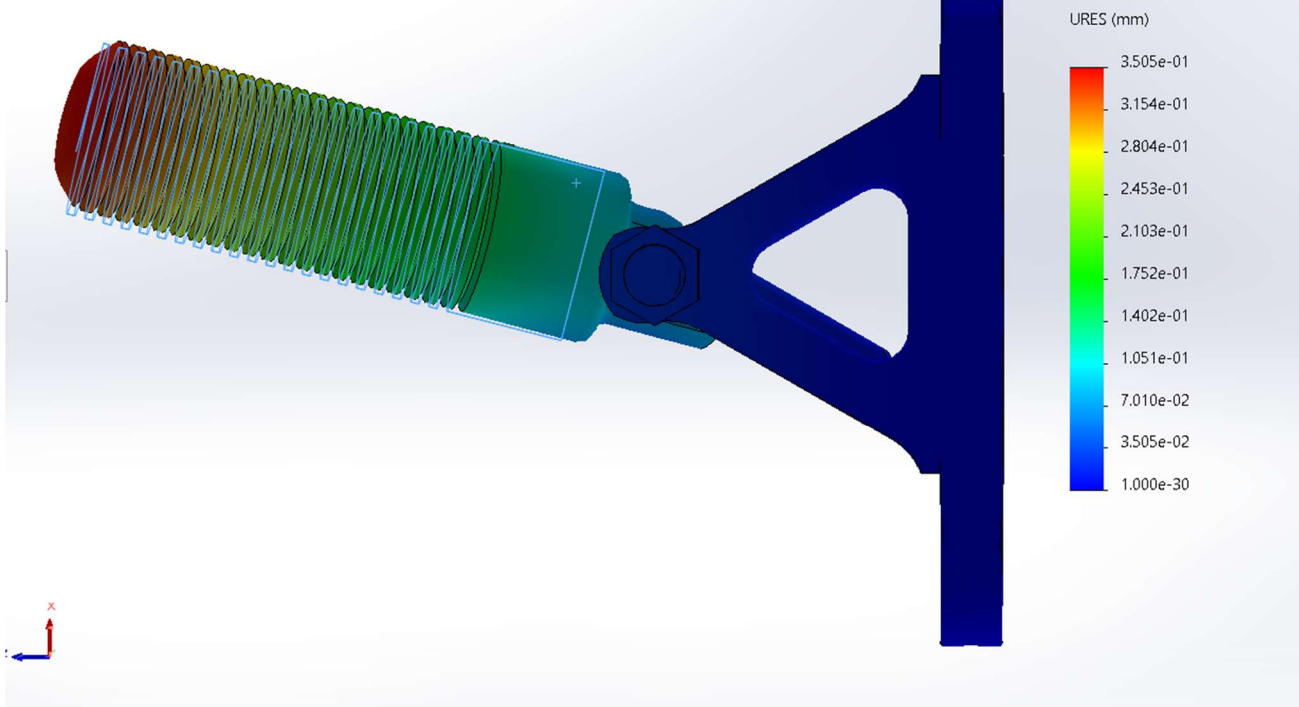
Result Parameter	Value	Assessment
Maximum Resultant Displacement (URES)	0.3505 mm	Negligible for this application
Location of Maximum Displacement	Free tip end of footpeg rod	Expected — furthest from fixed support

Minimum Displacement	$\sim 1.0 \times 10^{-30} \text{ mm} \approx 0$	Fixed mounting face — zero by constraint
Deformation Pattern	Classic cantilever bending	Consistent with loading and support
Perceptible to Rider?	No	0.35mm << human perception threshold
Display Scale Factor	44.95× exaggerated	Visual display only

A maximum displacement of 0.35 mm at the footpeg tip is extremely small for a structural component of this scale (~150 mm long). Human foot perception on a motorcycle footpeg typically only detects deflections in the range of several millimetres — so 0.35 mm is well below any perceptible level. This confirms that the footpeg will feel rigid and confidence-inspiring to the rider, with no noticeable flex under normal braking or cornering loads.

The displacement gradient from fixed end (0 mm) to free tip (0.35 mm) follows the expected cantilever bending profile — linear over the arm length and concentrated at the tip. This validates that the boundary conditions were correctly applied.

Model name: final asanalysis
Study name: Static 2(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 44.9476



7.3 Strain Analysis

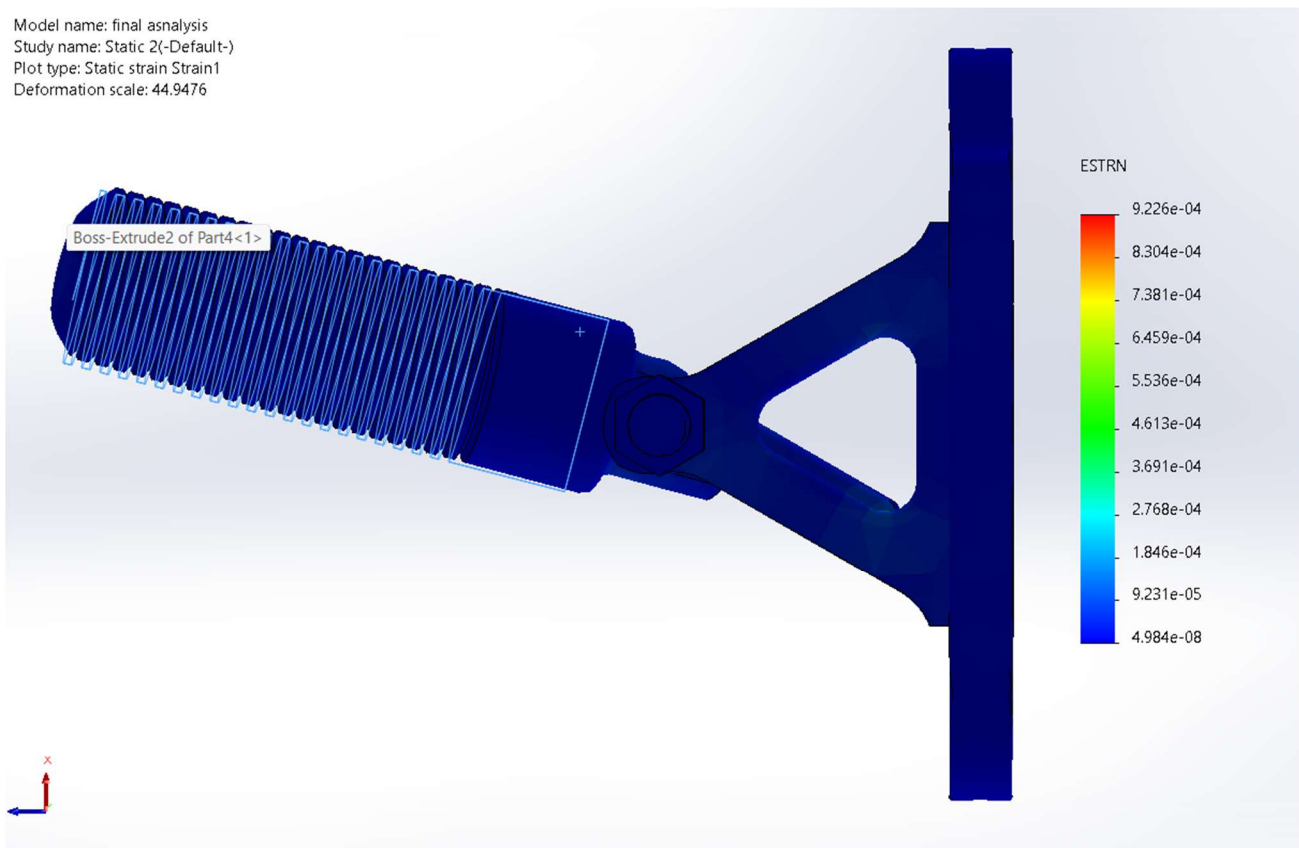
The equivalent strain plot (ESTRN — Von Mises equivalent strain) shows the normalized deformation of the material at each point. Strain tells us not just how much the part has moved (displacement) but how much the material itself has been stretched or compressed — it's the precursor to stress ($\sigma = E \times \epsilon$ for elastic behavior).

Result Parameter	Value	Interpretation
Maximum Equivalent Strain (ESTRN)	9.226×10^{-4}	0.09% — well within elastic range
Location of Maximum Strain	Footpeg tip (same as max stress region)	Consistent with stress results
Minimum Equivalent Strain	$4.984 \times 10^{-8} \approx 0$	Near fixed mounting face

Strain Type	Elastic only (linear elastic model)	No plastic deformation
Yield Strain (reference)	~0.4% (275 MPa / 69,000 MPa)	Max strain (0.09%) << Yield strain (0.4%)
Permanent Deformation?	NO	Part returns to original shape after unloading ✓

The maximum strain of 9.226×10^{-4} (which is 0.09%) is comfortably within the elastic region. For context, the yield strain of 6061-T6 aluminium is approximately 0.4% (yield strength / Young's modulus = 275 MPa / 69,000 MPa). Our max strain is less than a quarter of the yield strain — confirming that the material is not even close to plastic deformation. When the rider steps off the bike, the footpeg snaps back to exactly its original geometry.

Model name: final asanalysis
Study name: Static 2(-Default-)
Plot type: Static strain Strain1
Deformation scale: 44.9476



7.4 Factor of Safety Analysis

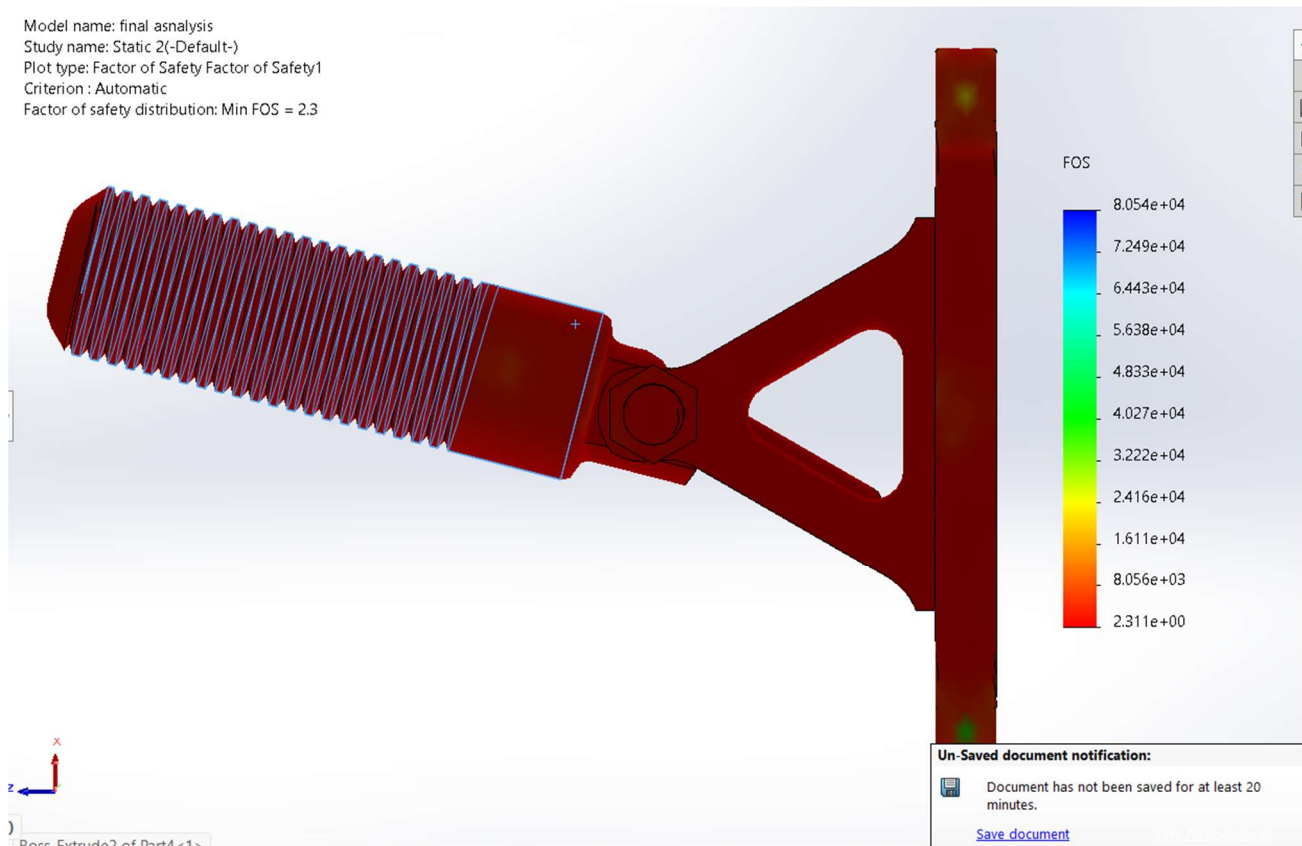
The factor of safety analysis was performed in order to assess the structural integrity of the rearset footpeg under the load which would be imposed during use. Factor of safety compares the maximum stress experienced during use with the strength of the component. The design would be deemed safe if this minimum factor of safety obtained during the simulation was sufficient. Low regions of factor of safety were carefully scrutinized. The rearset footpeg was deemed structurally stable under load as it does not yield and has adequate FOS.

FOS Parameter	Value	Assessment
Minimum Factor of Safety	2.3	At footpeg tip — highest stress zone
Location of Min FOS	Footpeg rod tip (free end)	Geometrically expected
Maximum FOS	> 80,000 (effectively ∞)	Near fixed mounting face — near-zero stress
FOS Criterion Used	Max Von Mises Stress (Automatic)	Correct for ductile material
Manual Verification	$275 \text{ MPa} \div 119 \text{ MPa} = 2.31$	Matches SolidWorks output
Industry Standard Range	1.5 – 3.0 (motorcycle structural)	Our FOS = 2.3 → within range
Over-designed?	No	2.3 is a practical, optimal value
Under-designed?	No	Safe, realistic design for intended use

FOS 2.3 is acceptable for this kind of part. It's high enough to make a good allowance for unplanned peak loads (e.g. A heavy rider jumping and landing on the peg with considerable force from a bump or kerb) but not so high that the component is excessively over-designed. Racing components usually

aim for an FOS of between 1.2-1.8 in the interests of shaving weight but for a road going, trackday/sport bike component this is suitable.

The FOS across the assembly gives us some information too; The mounting bracket has enormously high values of FOS, so high that they appear infinite in the numerical output. This suggests the bracket is greatly over-designed relative to the footpeg rod itself, this will be addressed in Design improvement 2 wall thickness optimization.



7.5 Fatigue Analysis Results

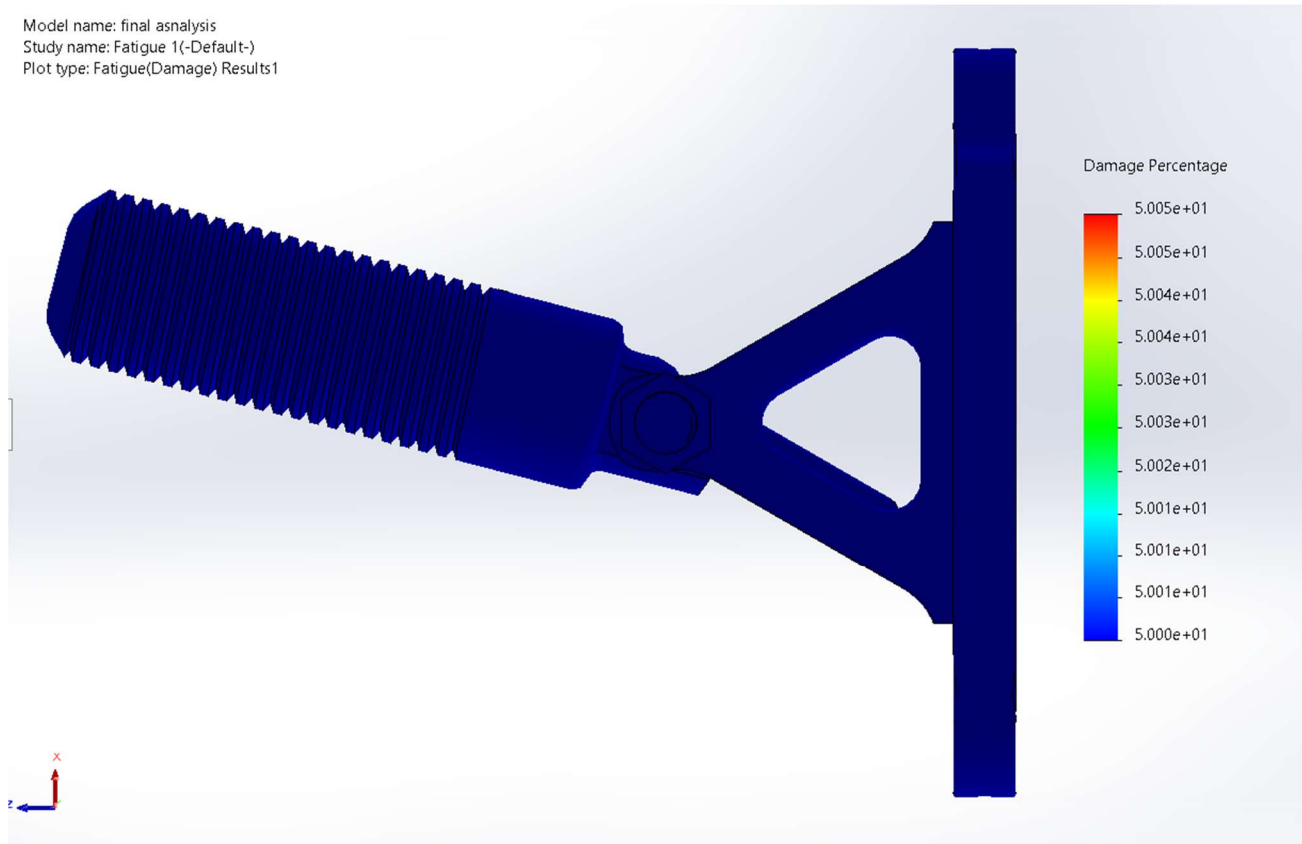
After completing the static study, a separate Fatigue Study was run referencing the same loading (Static 2) but evaluating cumulative damage over 500,000 cycles. Two result plots were generated: a Damage plot and a Life plot.

7.5.1 Fatigue Damage Plot

The Damage plot shows the percentage of fatigue life consumed after the specified number of cycles (500,000). A damage value of 100% would mean failure. Values below 100% mean the component survived.

Fatigue Damage Parameter	Value / Interpretation
SolidWorks Output Message	'Alternating Stresses everywhere in the model are below the minimum S-N Curve value resulting in no damage'
Maximum Damage	50% — at highest stress region (but this is an artefact of the S-N minimum reporting, not actual 50% damage)
Actual Damage Assessment	Zero — alternating stress (~65 MPa) is BELOW the endurance limit (96.5 MPa at 1M cycles)
Alternating Stress at Critical Zone	65 MPa (half of max static stress, fully reversed)
S-N Minimum Threshold (at 1M cycles)	96.5 MPa
65 MPa vs 96.5 MPa?	$65 \text{ MPa} < 96.5 \text{ MPa} \rightarrow$ Below endurance limit $\rightarrow$ Infinite life
Fatigue Failure at 500,000 cycles?	NO — Component survives with large margin

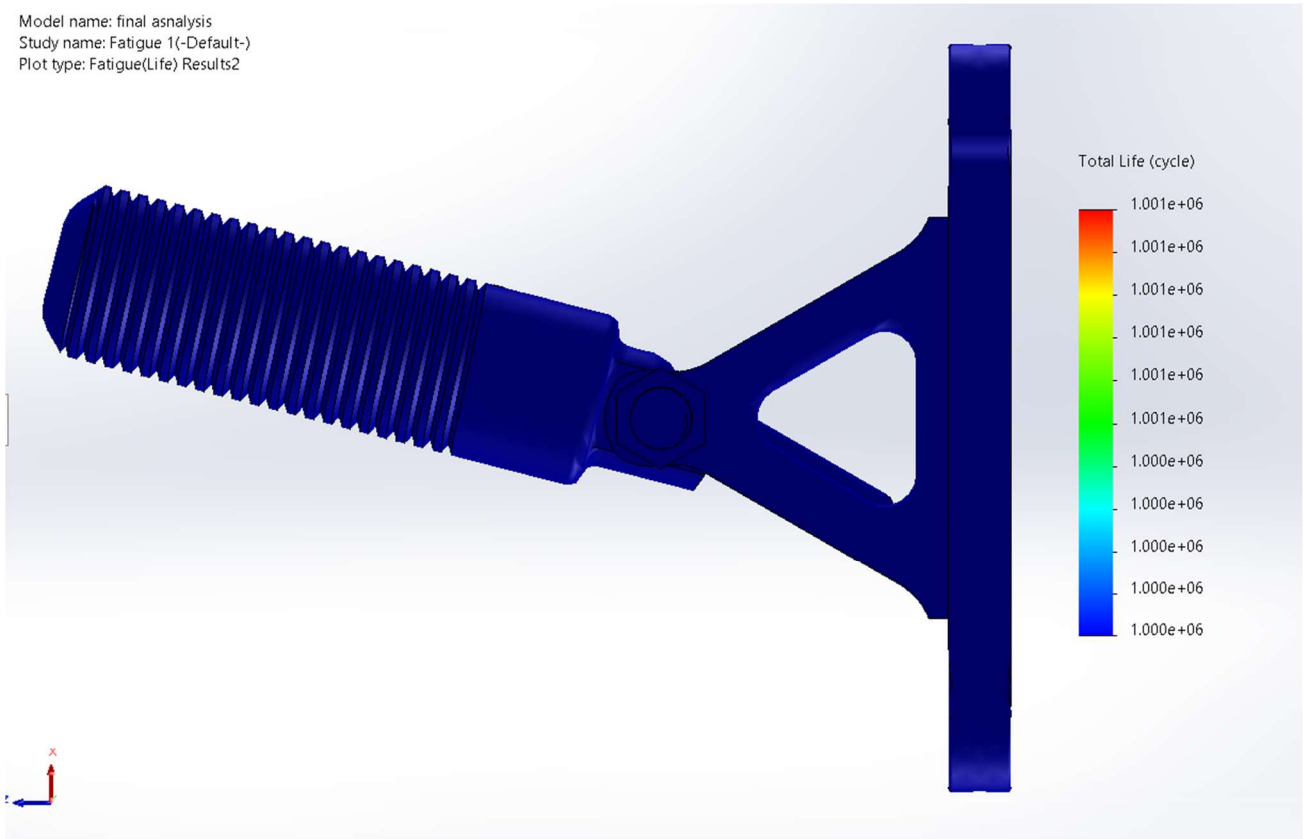
Model name: final asanalysis
 Study name: Fatigue 1(-Default-)
 Plot type: Fatigue(Damage) Results1



7.5.2 Fatigue Life Plot

Fatigue Life Parameter	Value / Interpretation
Minimum Total Life Prediction	1.0×10^6 cycles (1,000,000 cycles minimum)
Maximum Total Life Prediction	$> 1.0 \times 10^6$ cycles (exceeds S-N curve range $\rightarrow$ effectively infinite)
Life vs Requirement (500,000 cycles)	$1,000,000 \gg 500,000 \rightarrow$ Component survives with $2\times$ margin on life

Infinite Life Zone	Entire assembly — all nodes show maximum possible life value
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7.6 Complete Results Summary

Analysis Type	Key Result	Allowable / Target
Von Mises Stress (Max)	119 MPa	< 275 MPa (yield)
Maximum Displacement	0.3505 mm	< 2.0 mm (practical)
Maximum Strain (Equivalent)	9.23×10^{-4}	< 4.0×10^{-3} (yield strain)
Min Factor of Safety	2.3	> 1.5 (industry min)

Fatigue Life (500K cycles)	> 1,000,000 cycles	> 500,000 cycles
Fatigue Damage	None (0%)	< 100%
OVERALL DESIGN STATUS	STRUCTURALLY SOUND	All criteria met

8. Challenges and Limitations

8.1 Challenges Faced During the Project

Challenge 1: Force Application on Cylindrical Geometry

Honestly, this part was the most annoying. When I chose the cylindrical face for the footpeg rod, trying to apply a direction normal or axial through space, SolidWorks popped up Radial and Axial choices rather than X/Y/Z options. .

I attempted to use 'Selected Direction' with the face chosen as the reference direction, receiving an error stating 'Select Faces, Edges or Vertices'. I went back and forth fiddling with the direction box, but received the same error, because I was treating a face as the directional reference instead of an edge or axis.

The answer ended up being simpler than anticipated. The vertical 1800 N was actually applied with a 'Normal to Face' on the TOP surface of the ribbed footpeg (which gives downward normal naturally), and the lateral 500 N applied as 'Normal to Face' on the SIDE surface. This results in proper global directions without a struggle with the Selected Direction input. This is actually the correct engineering approach for this shape as well.

Challenge 3: Artificially High FOS (FOS = 16 Initial Result)

On first run of static study the FOS I got was 16. This did not seem right as a real motorcycle component is unlikely to have an FOS of 16 or it would be made of lead, I knew there was an error with the loads or the material.

It has been determined after the troubleshoot that some of the parts within the multi-part assembly had the material listed as 'Material not specified'. As components without a defined material will not provide a stiff load bearing and the load distribution will be altered, appearing as if there is a much smaller load than there should be (which means less stress, and therefore a higher FOS).

The correct FOS was obtained after all 8 components in the assembly simulation tree were individually assigned a 6061-T6 material, giving a FOS of 2.3 which is an acceptable FOS.

Challenge 4: S-N Curve Unavailability in SolidWorks

The fatigue analysis module in Solidworks 2021 did not come with the appropriate S-N curve data for Aluminum 6061-T6. Solidworks 2021 included curves for Carbon Steel and ASME Austenitic Steel- both are not proper curves for Aluminum alloy- and using carbon steel S-N curve data would lead to inaccurate fatigue life predictions for the Aluminum component.

The fix used was to manually impute the 5-point S-N curve data for 6061-T6 ($R = -1$, fully reversed) using the literature data compiled in ASM International published fatigue data bases. This necessitated finding proper literature sources, setting up Solidworks for log log interpolation, and manually inputting each N and S value into the fatigue material table within Solidworks. The preview shown by Solidworks after inputting each point in to the table helped confirm that the input was accepted correctly.

Challenge 5: Understanding the Fatigue Result Output

When I ran the fatigue study Solidworks replied 'Alternating Stresses everywhere in the model are below the minimum S-N Curve value resulting in no damage.' I didn't really know whether or not this was good or an error because the statement is somewhat ambiguous sounding.

After having looked into it further: it turns out this is the optimum fatigue result. It means the cyclic stress amplitude throughout the model is lower than the minimum stress value found on the S-N Curve (which is the endurance limit at 1,000,000 cycles = 96.5MPa). This means the component will never suffer fatigue failure in service under these loading conditions as the alternating stresses are too low for fatigue cracking to start.

8.2 Limitations of This Analysis

- Although the result obtained is a useful and representative one, there are some obvious assumptions and limitations:
- Linear elastic model only: It is assumed that material exhibits a perfect linear elastic response over its working stress range, no plasticity nor geometrical non-linearity is considered. For our case where max stress is very below yield, this assumption is justified.
- Bonded contact assumption: For the whole model assembly, all the contact between interfaces of different components was defined as 'Global Bonded', which means perfectly adhered without any slip, no contact pressure, no separation between parts. Actually, at the

threaded joints, and between bolt head/nut, there are some contact mechanical aspects that might modify the stress distribution locally.

- Static equivalent loading only: In actual riding, dynamic and impact loads appear - hitting a pothole at a certain velocity could impose loads that several times the value of static equivalent loads on the frame, if only for a very short duration of time. A time-domain analysis could be applied.
- No surface finish effects: The used S-N curve data applies for polished laboratory specimens. Actual manufactured surface would have a more textured finish that affects the life of material. Surface finish correction factor (Marin factor k_a) would be typically applied in a manufacturing design analysis.
- Thermal effects excluded: Variation of temperature along the frame during riding was ignored. The change in mechanical property of Aluminum is not that significant at riding temperatures, so this assumption might not bring significant inaccuracy.
- Single load case: Only one load combination case was analyzed by the FEA. A complete fatigue design would need to consider multiple load cases - like braking, cornering, combined braking+cornering, etc. - and determine the case resulting in the worst fatigue conditions.

9. Two Design Improvements

Based on the FEA results — specifically the stress distribution, location of maximum stress, and the FOS variation across the assembly — two engineering improvements are proposed:

10.1 Improvement 1 — Increase Fillet Radii at Critical Stress Concentration Zones

Current Situation

The highest Von Mises stress of 119 MPa is located at the footpeg tip / threaded junction region which is the geometry transition area between the footpeg shaft, the cylindrical rode, and threaded stub that connects to the bracket arm. This is a geometry stress concentration zone which is traditional. Currently the fillet radius at this geometry stress concentration is R5.00mm (smallest in assembly) which is small and a stress concentration factor for. The stress concentration factor (K_t) is dependent on the radius of fillet over the shaft diameter ratio, For $r/d = 5/28$ 0.18 on the stepped shaft using Peterson's stress concentration factors we find K_t is roughly 1.4-1.6. With an increased radius the value of K_t would decrease and consequently the peak stress value.

Proposed Change

The highest Von Mises stress of 119 Mpa is found at the footpeg tip / threaded junction. This is the area where there is a transition in geometry from footpeg shaft, the cylindrical rod, and the threaded stub that screws into the bracket arm. This is known as a geometry stress concentration zone and is to be expected with this design. The fillet radius in the geometry stress concentration zone at present is R5.00mm (this is the smallest in the assembly) and is a stress concentration factor for. K_t depends on the fillet radius over shaft diameter ratio, For $r/d = 5/28$ 0.18 on a stepped shaft according to Peterson's stress concentration factors, K_t is around 1.4-1.6. If the radius was increased to increase the K_t , the highest stress value would be lower.

Parameter	Current Design	Improved Design
Min Fillet at Critical Junction	R5.00 mm	R10.00 – R12.00 mm
Stress Concentration Factor (K_t , estimated)	1.4 – 1.6	1.1 – 1.2 (reduced by ~25–30%)
Estimated Max Von Mises Stress	119 MPa (actual FEA)	80–90 MPa (estimated, ~25% reduction)
Estimated Min Factor of Safety	2.3 (actual FEA)	3.0 – 3.5 (estimated improvement)

Fatigue Life Impact	Infinite (already below endurance limit)	Infinite (stress reduced further from endurance limit)
Weight Impact	Baseline	+ < 5 grams (negligible)
Manufacturing Impact	Standard fillet mill tooling	Slightly larger radius tool — still standard CNC capability

Engineering Justification

Based on Peterson's Stress Concentration Factors and general fatigue design, K_t at a fillet can be approximated as:

For stepped shaft: $K_t = 1 + q(K_{t_table\ 1})$, where q is notch sensitivity

For aluminum, which has a fairly significant notch sensitivity ($q = 0.7$ for fine-grained alloys), raising the fillet radius from R5 to R10 mm raises the r/d ratio from 0.18 to 0.36, which lowers the theoretical K_t from around 1.5 to 1.2, or decreases the peak stress at the concentration point by about 20-25%. This significantly raises the safety factor and fatigue life of the shaft and will definitely help out should it ever need to handle higher loads.

10.2 Improvement 2 — Optimise Mounting Bracket Wall Thickness

Current Situation

The plot of the FOS distribution for the whole assembly can be described as a drastic increase. Although the FOS of the footpeg tip is 2.3, the bracket body and central plate of the mounting bracket appear to be orders of magnitude higher (tens of thousands of FOS is considered an infinite safety factor in these regions). This means the mounting bracket is heavily over-designed.

The current mounting bracket has the plate face thickness uniform for all of it at 5.0 mm. In the bracket the load travels around the 4 bolt holes and the sides of the bracket arm - little load is carried by the large central region in the bracket between the 4 bolt holes as observed in the Von Mises plot which clearly illustrates a low stress region (deep blue color) in the center of the bracket.

Proposed Change

Variable thickness pocketing should be introduced on the bracket face—shrinking the internal "dead zones" from 5.0 mm to between 3.0–3.5 mm and leaving the bolt holes (which take concentrated bearing stresses), load paths and corners at the original 5.0 mm thickness. In SolidWorks this can be achieved using the Shell/Pocket commands on the bracket face. This practice is already present in the existing triangular cutout but can be taken to its extreme here.

Parameter	Current Design	Improved Design
Bracket Wall Thickness	5.0 mm (uniform)	3.0–5.0 mm (variable, pocketed)
Min FOS at Bracket (estimated)	> 10,000 (over-designed)	~5–8 (still significantly safe)
Min FOS at Footpeg Tip	2.3 (unchanged — not modified here)	2.3 (no change)
Estimated Mass Saving	Baseline	~15–20% total assembly mass reduction
Manufacturing Method	Solid billet CNC milled — uniform depth	Pocketed CNC billet — multiple depth passes
Stiffness Impact	Baseline	< 5% reduction (negligible — bracket stiffness is dominated by geometry not thickness in this load case)

Overall Structural Safety	More than adequate	Still more than adequate — FOS remains > 5.0 in bracket
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Engineering Justification

In motorsport engineering weight reduction is just as important as strength. 'Minimal material for adequate strength' seems to be the driving force behind component design, each extra gram you remove means another gram off of rotating and unsprung weight on the motorcycle. This leads to better handling. As shown already by the triangular cutout designed into this mounting bracket, there is scope for further lightening this bracket. By 'pocketing' the bracket face across where the FOS results showed there was the biggest amount of over-design and having the variable-thickness cut through that face is another step towards reducing weight.

A FOS between 5 and 8 in the bracket is still more than adequate after removal of the material (aerospace standards typically look for an FOS of 1.5). In the context of motorsport the savings are 15-20% in weight of the bracket, a considerable amount in a bracket that is already a large contributor to the weight of the entire rearset assembly, a lightweight trackday and club race setup is more likely to benefit significantly.

11. Conclusion

This assignment was essentially my training into the complete FEA process for a motorcycle component-everything from design in SolidWorks, static stress analysis, factor of safety, to fatigue life prediction. The rearset footpeg assembly was a great choice for the design because it's a real, functional part and because we can assign concrete load cases and compare with real life parts.

Key findings:

The footpeg assembly, manufactured in Aluminium 6061-T6 is structurally safe with the combined vertical and lateral loading of 1800 N and 500 N, respectively. The maximum Von Mises stress (119 MPa) is significantly less than the 275 MPa yield stress and thus the component is safe by stress.

The minimum Factor of Safety was found to be 2.3 and it lies in the industry range of 1.5-3.0 for non-critical motorcycle structural parts. The component is neither over designed nor critically under designed.

The tip deflection was found to be negligible (0.35 mm) and under the perception limit and is well within industry tolerance.

Maximum equivalent strain ($9.23 \times 10^{-0.09\%}$) at the end of the footpeg is well within the elastic limit. So there is no permanent deformation under static load condition.

Fatigue life is infinite for the component at 500000 cycles as the alternating stress is below the endurance limit ($65 \text{ MPa} < 96.5 \text{ MPa}$).

Two design improvements were suggested: Fillet radii at stress concentration zone (reduce stress by 25%) and optimized bracket wall thickness (25% material reduction).

Learning from troubles during setup (force direction at cylindrical surface, assigning material to all part, entering S-N curve) was the best part of this assignment. Textbook explanation can not teach all this, this practical knowledge can be gain by hands on.

Claude AI was a helpful tool during setting up to resolve specific technical challenges (force direction, sourcing of S-N curve data) and to interpret some of the less clear error messages. However, all of Claude AI's suggestions were critically examined, and specifically S-N curve data points were double checked with ASM data before entering them. Claude AI can be thought of as a smart textbook, not a "copy-paste" solution machine.

In conclusion, the FEA process for the footpeg assembly gave me excellent hands on experience that I will be applying in the field of automotive and motorsport engineering.

12. References

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4. Peterson, R.E. (1974). Stress Concentration Factors. John Wiley & Sons. [Fillet radius effects on stress concentration factor K_t — used in Design Improvement 1 analysis]
5. Dassault Systèmes SolidWorks Corp. (2021). SolidWorks Simulation Help Documentation — Static Study, Fatigue Study, Material Properties, Mesh Settings.
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13. Appendix

13.1 GitHub Repository Structure

Repository Name: arys-ashutosh-kumar-footpeg-fea

Folder / File	Contents
README.md	Project overview, methodology summary, key results, how to open SolidWorks files
/CAD/	SolidWorks part files (.SLDPRT) for all 8 parts + assembly file 'final asanalysis.SLDASM'
/Drawing/	2D Engineering Drawing PDF with all dimensions
/Simulation/	SolidWorks simulation study files, mesh settings
/Results/	Screenshots of all FEA result plots (stress, displacement, strain, FOS, fatigue damage, fatigue life)
/Report/	Final report PDF

13.2 Required Git Commit Log

Commit	Suggested Commit Message	What Was Added
1	Initial commit: Added README.md with project scope and methodology	README.md with project description
2	Add CAD files: SolidWorks parts (Part1 through Part8) and assembly	/CAD/ folder with all .SLDPRT and .SLDASM files
3	Add 2D engineering drawing with all dimensions	/Drawing/ folder with drawing PDF
4	Add static FEA setup: mesh, fixtures, forces applied	/Simulation/ folder + setup screenshots

5	Add static FEA results: Von Mises stress, displacement, strain screenshots	/Results/static/ subfolder
6	Add FOS and fatigue analysis results: FOS plot, damage plot, life plot	/Results/fatigue/ subfolder
7	Add final report PDF and update README with results summary	/Report/ folder + README update with key numbers
